

Home Valuation Model Spec v1

Purpose

Build a production-grade home valuation engine for an app that estimates a property's current fair value, uncertainty range, and forecast path using a layered probabilistic model. The engine must support three user-facing forecast modes: Conservative, Moderate, and Optimistic.

The model is designed to outperform simple comp averaging and naive appreciation roll-forwards by combining:

- property-level structural features
 - address-level spatial effects
 - neighborhood regime dynamics
 - macro housing state
 - property-specific upgrade / deterioration state
 - probabilistic uncertainty and scenario simulation
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Product Requirements

Core user-facing outputs

For any property address, the system should return: 1. Current fair value estimate 2. Fair value range (50%, 80%, and 95% confidence bands) 3. Conservative / Moderate / Optimistic scenario values 4. Attribution breakdown 5. Forecast path for 3, 6, 12, 24, 36, and 60 months 6. Comparable sales summary 7. Confidence score 8. Explanation layer in plain English

Internal model goals

The model should:

- separate structural value from market timing
 - capture micro-location effects below the zip code level
 - detect whether a neighborhood is stable, improving, accelerating, overheated, or correcting
 - ingest noisy observations from sales, listings, AVMs, tax assessments, and permits
 - simulate future value under multiple regimes rather than using a single deterministic CAGR
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Modeling Philosophy

The model should not treat a home like a continuously traded stock. Housing is:

- illiquid

- sticky
- spatially dependent
- heterogenous
- exposed to regime shifts
- observed only intermittently

Therefore, the core architecture should be a **hierarchical spatial state-space model with regime-switching neighborhood dynamics**.

This yields a system that is advanced but still interpretable.

Mathematical Core

Let:

- i = property index
- t = monthly time index
- $g(i)$ = neighborhood group of property i
- ℓ_i = geographic coordinates / spatial embedding of property i
- $x_{i,t} = \log V_{i,t}$ = latent log fair value of property i at time t

Latent value decomposition

$$x_{i,t} = b_i + s(\ell_i) + u_{g(i),t} + m_t + h_{i,t}$$

Where:

- b_i : structural baseline from the home itself
 - $s(\ell_i)$: spatial micro-location effect
 - $u_{g(i),t}$: evolving neighborhood state
 - m_t : macro housing state
 - $h_{i,t}$: property-specific evolving state
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Layer 1: Structural Baseline

Definition

The structural baseline represents the intrinsic value of the house independent of the current cycle.

$$b_i = \alpha + f_{\theta}(X_i)$$

Required inputs

Property feature vector X_i :

- living area
- lot size
- beds
- baths
- year built
- effective age
- stories
- garage spaces
- pool
- HOA
- waterfront / water view
- cul-de-sac
- corner lot
- gated community
- ADU / guest house / flex space
- zoning class
- school assignments
- construction quality
- renovation quality
- roof age
- HVAC age
- kitchen / bath update level
- solar status
- permit-based additions

Modeling notes

Use nonlinear effects. Recommended options:

- spline-based GAM
- monotonic gradient boosting for research comparison
- Bayesian additive model for production explainability

Why nonlinear

Examples:

- bath count value saturates
 - lot size premium is curved
 - year built matters differently in 1955 vs 2005 stock
 - pool value depends on climate and price tier
 - renovations create stepwise uplift rather than smooth linear change
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Layer 2: Spatial Micro-Location

Definition

This layer captures the value of the exact address and its immediate surroundings beyond zip code or tract averages.

$$s(\ell_i)$$

Recommended implementation

Research version:

- Gaussian process over latitude / longitude

Production version:

- graph-based spatial smoother over parcel / block / tract adjacency
- optionally enriched with a learned spatial embedding

Required location features

- latitude / longitude
- road class adjacency
- proximity to highways
- proximity to power lines / rail / airports
- distance to parks
- distance to water
- flood zone
- wildfire / hazard zone where applicable
- walkability metrics
- retail access
- grocery / hospital / school access
- transit access
- travel time to job centers
- school boundary quality
- block-face similarity cluster

Important note

Prefer network travel distance over straight-line distance for practical amenity access.

Layer 3: Macro Housing State

Definition

This layer captures market-wide financing and demand conditions.

$$m_t = Am_{t-1} + Bz_t + \eta_t$$

Required macro inputs

Monthly or quarterly:

- 30-year mortgage rate
- local inventory / months of supply
- median DOM
- regional unemployment
- wage growth
- CPI / inflation
- housing permits
- migration flow proxy
- builder sentiment / construction activity if available
- consumer sentiment if available
- local rent inflation

Recommended modeling options

- dynamic factor model
- low-dimensional state-space VAR

Output interpretation

This layer should answer:

- is financing supportive or restrictive?
- is supply expanding or tight?
- is market demand stable, cooling, or recovering?

Layer 4: Neighborhood Regime / Transition Layer

Definition

This layer models local neighborhood transition, redevelopment, and possible gentrification as a latent regime process.

Regime states

Recommended set:

- stagnant
- stable
- improving
- accelerating
- overheated
- correcting

Let:

$$r_{g,t} \in \{1, 2, 3, 4, 5, 6\}$$

Regime transition model

$$\Pr(r_{g,t} = k \mid r_{g,t-1}, q_{g,t}) = \text{softmax}(a_k + b_k^\top q_{g,t})$$

Neighborhood state evolution

$$u_{g,t} = u_{g,t-1} + \delta_{r_{g,t}} + \gamma^\top q_{g,t} + \omega_{g,t}$$

Leading indicators vector $q_{g,t}$

- permit count trend
- permit dollar volume trend
- rezoning / land-use approvals
- infrastructure announcements
- school trend
- crime trend
- rent growth vs nearby areas
- investor acquisition share
- flip / renovation intensity
- new retail tenant quality
- DOM compression
- sale-to-list ratio improvement
- income growth
- educational attainment shift
- code enforcement cleanup / blight reduction where available

Design principle

Do not expose “gentrification” as a blunt scalar. Expose it internally as a transition probability and a neighborhood regime score.

Layer 5: Property-Specific Evolution

Definition

Captures changes specific to one property over time.

$$h_{i,t} = \rho h_{i,t-1} + \kappa^\top R_{i,t} - \lambda^\top D_{i,t} + \xi_{i,t}$$

Positive state inputs $R_{i,t}$

- major remodel permits
- new roof
- HVAC replacement
- kitchen remodel
- bathroom remodel
- energy upgrades
- pool addition
- ADU / enclosed patio / office conversion where permitted

Negative state inputs $D_{i,t}$

- aging without upgrades
- deferred maintenance flags
- storm / water damage
- insurance risk repricing
- external obsolescence
- condition downgrades from listing text or image analysis

Operational note

In v1, this can be approximated through permit data + listing-history change detection + manual user adjustments.

Observation Model

The latent fair value is not directly observed. The app ingests multiple noisy proxies.

Sale observation

$$y_{i,t}^{sale} = x_{i,t} + c^\top w_{i,t} + \varepsilon_{i,t}$$

Where $w_{i,t}$ includes:

- urgency / distress flags
- seasonal selling effect
- concession environment

- bidding intensity
- off-market / weak-marketing penalty

Use fat-tailed error:

$$\varepsilon_{i,t} \sim t_{\nu}(0, \sigma_{r_{g,t}})$$

Listing observation

$$y_{i,t}^{list} = x_{i,t} + \mu_{i,t}^{markup} + \varepsilon_{i,t}^{list}$$

Tax observation

$$y_{i,t}^{tax} = x_{i,t} + b_{g,t}^{county} + \varepsilon_{i,t}^{tax}$$

Third-party AVM observation

$$y_{i,t,s}^{avm} = x_{i,t} + b_{s,t}^{avm} + \varepsilon_{i,t,s}^{avm}$$

Where s indexes AVM source such as Zillow, Redfin, Realtor, or Homes.com.

Why multi-source observations matter

- sales are sparse
- listings are aspirational
- tax values lag
- AVMs vary in bias
- ensemble observations stabilize the estimate

Sale Selection Correction

Homes that sell are not random. A correct model should account for sale propensity.

$$\Pr(S_{i,t} = 1) = \text{logit}^{-1}(a + d^{\top} z_{i,t})$$

Where $z_{i,t}$ may include:

- length of ownership
- rate-lock incentive
- equity level
- seasonality
- neighborhood turnover
- life-event proxies if available

In v1 this can be omitted if data is sparse, but the architecture should reserve a place for it.

User-Facing Scenario Models

The app should expose three forecast modes using the same core model and different assumptions over macro, neighborhood regime, and local comp realization.

1. Conservative

Use when the market is restrictive or the user wants downside-aware planning.

Assumptions:

- lower macro drift
- higher financing drag
- neighborhood state stays stable or softens
- lower realization of upside from development signals
- lower percentile of plausible comp outcome

Implementation:

- use 20th to 30th percentile posterior path
- reduce neighborhood drift multiplier
- increase risk discount / widen downside tail

2. Moderate

Base case.

Assumptions:

- current macro consensus persists
- local comps anchor value
- neighborhood remains in current regime unless strong evidence says otherwise
- neutral realization of development pipeline

Implementation:

- use posterior median or 50th percentile path

3. Optimistic

Use for upside exploration, not marketing hype.

Assumptions:

- financing conditions improve or stabilize favorably
- local demand remains healthy
- neighborhood transition indicators realize positively

- comps clear toward the upper credible band

Implementation:

- use 70th to 80th percentile posterior path
- allow higher but still plausible neighborhood drift
- tighten discount for current buyer hesitation if indicators support it

Important rule

These are not three separate models. They are three calibrated views over the same posterior distribution.

Confidence and Quality Scores

The engine should return metadata alongside valuation.

Confidence score inputs

- number of recent relevant comps
- distance-weighted comp quality
- observation agreement across AVMs / listings / tax / sale data
- staleness of most recent local sales
- neighborhood regime certainty
- missingness in property features

Recommended confidence bands

- 95: extremely high quality estimate
 - 80: strong estimate
 - 60: usable but uncertain
 - below 60: display caution
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Data Sources Required

Essential

- county property appraiser / assessor data
- transaction history
- active / pending / sold listing data
- geospatial parcel data
- macro market data
- mortgage rate time series

Strongly recommended

- permit data
- crime data
- school rating trend
- flood / hazard maps
- walkability / amenity access
- local rent data
- infrastructure / planning feeds

Optional / advanced

- image-based condition scoring from listings
 - NLP from listing descriptions
 - satellite / street-view context embeddings
 - investor acquisition signals
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Feature Schema

Property table

- property_id
- address_full
- latitude
- longitude
- beds
- baths
- sqft_living
- sqft_lot
- year_built
- effective_year_built
- stories
- garage_spaces
- pool_flag
- hoa_flag
- water_flag
- culdesac_flag
- corner_lot_flag
- zoning_code
- assessed_value
- tax_year
- permit_summary
- last_sale_price
- last_sale_date

Neighborhood monthly table

- geography_id
- month
- permit_count
- permit_value
- inventory_months
- dom_median
- sale_to_list_ratio
- crime_index
- school_trend_score
- rent_growth
- investor_share
- retail_quality_score
- infrastructure_score
- regime_prior

Macro monthly table

- month
- mortgage_rate_30y
- unemployment_rate
- wage_growth
- cpi
- housing_permits
- migration_proxy
- metro_inventory
- metro_dom

Observation table

- property_id
- month
- observation_type
- observation_value_log
- source
- reliability_weight
- transaction_flags

Service Design for the App

1. Address ingestion service

Responsibilities:

- normalize address

- geocode
- attach parcel / assessor record
- fetch transaction history
- fetch nearby comps

2. Feature engineering service

Responsibilities:

- compute structural features
- spatial features
- neighborhood features
- macro joins
- comp relevance scores

3. Current value engine

Responsibilities:

- estimate present latent fair value
- aggregate observation sources
- generate value band and attribution

4. Regime engine

Responsibilities:

- estimate current neighborhood regime probabilities
- score development / planning / transition strength
- adjust local drift and volatility

5. Forecast engine

Responsibilities:

- simulate future macro / neighborhood / property states
- return Conservative / Moderate / Optimistic paths
- return confidence intervals

6. Explanation service

Responsibilities:

- translate attribution into plain English
- explain why this home differs from citywide averages
- explain what pushed the conservative and optimistic scenarios apart

Suggested API Contract

Request

```
{
  "address": "207 Cape Sable Dr, Orlando, FL 32825",
  "horizon_months": 60,
  "include_scenarios": true,
  "include_attribution": true,
  "user_adjustments": {
    "renovation_level": "none",
    "condition_override": null
  }
}
```

Response

```
{
  "property": {
    "address": "207 Cape Sable Dr, Orlando, FL 32825",
    "property_id": "...",
    "beds": 5,
    "baths": 3,
    "sqft_living": 2713
  },
  "current_value": {
    "fair_value": 535000,
    "range_50": [529000, 541000],
    "range_80": [525000, 550000],
    "range_95": [510000, 565000],
    "confidence_score": 78
  },
  "scenarios": {
    "conservative": 525000,
    "moderate": 535000,
    "optimistic": 550000
  },
  "attribution": {
    "structure": 0.42,
    "micro_location": 0.18,
    "macro": -0.06,
    "neighborhood": 0.09,
    "property_specific": 0.02,
    "comp_residual": 0.35
  },
}
```

```
"regime": {
  "stable": 0.46,
  "improving": 0.24,
  "correcting": 0.18,
  "overheated": 0.04
},
"forecast": {
  "12_months": {
    "conservative": 520000,
    "moderate": 540000,
    "optimistic": 565000
  }
},
"explanation": "Current value is driven primarily by recent nearby large-home comps. Macro financing conditions remain a drag, which is why the estimate sits below a simple Orlando metro appreciation roll-forward."
}
```

Training and Backtesting

Walk-forward protocol

For each backtest date t : 1. train only on data available by t 2. estimate current latent fair values 3. forecast 3, 6, 12, and 24 months ahead 4. compare against realized sales

Metrics

- MAE / RMSE on realized sale prices
- calibration of 50 / 80 / 95 bands
- median absolute percentage error
- coverage of scenario bands
- neighborhood ranking accuracy
- attribution stability over time

Baselines to compare against

- metro HPI roll-forward
 - plain hedonic regression
 - comp average only
 - external AVM ensemble
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MVP vs Full Version

MVP

Use:

- structural features
- nearby comps
- spatial smoothing
- macro adjustment
- simple neighborhood score instead of full regime switching
- scenario values from posterior quantiles

Full v2

Add:

- formal regime switching
 - permit / planning feeds
 - property-specific evolution state
 - selection correction
 - image and NLP condition scoring
 - particle filtering for state updates
-

Recommended Build Order

Phase 1

- address ingestion
- comp retrieval
- structural model
- current fair value estimate
- confidence score

Phase 2

- macro state layer
- neighborhood score layer
- scenario outputs
- forecast engine

Phase 3

- regime probabilities
- permit / planning integration
- property update tracking

- uncertainty calibration

Phase 4

- full Bayesian refit pipeline
 - explainability dashboards
 - continuous backtesting monitor
-

Final Design Principle

The most important modeling decision is this:

Do not let the app reduce a property to “city appreciation + comp average.”

The engine should estimate:

- what the house is
- where the house is
- what the local neighborhood is becoming
- what the macro market is doing
- and how uncertain all of that is

That is what makes the valuation intelligent rather than merely automated.